

Infrastructure Risk Assessment — 2026

(CONFIDENTIAL)

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This risk assessment evaluates NovaTech's infrastructure risks as of Q2 2026. Top risks: (1) single-region EKS dependency — DR cluster in eu-west-1 is not yet at parity; (2) Qdrant cluster at 78% capacity — needs expansion before Q3 peak; (3) LLM vendor concentration (100% Anthropic) — no fallback model configured. Post-March 2026 incident, the security posture has improved but 3 audit findings remain open.

1. Risk Register

ID	Risk	Likelihood	Impact	Rating
R01	Single-region EKS — DR at 60% parity	Medium	Critical	High
R02	Qdrant at 78% capacity	High	High	High
R03	LLM vendor concentration (Anthropic)	Low	Critical	Medium
R04	3 open audit findings (A03/A06/A07)	Medium	Medium	Medium
R05	AWS cost overrun — GPU instances	High	Medium	Medium
R06	Key-person dependency: Deepa Iyer	Low	High	Medium

2. Qdrant Capacity Risk

The production Qdrant cluster is projected to reach 90% capacity by 2026-07-01 based on current ingestion rates (120 000 new chunks/week). At 90% utilisation, HNSW search performance degrades by approximately 40% due to increased disk I/O. A 3-node cluster expansion (r6g.2xlarge) is budgeted at \$4 200/month and has been approved in the Q3 engineering budget. Target expansion date: 2026-06-15.

3. LLM Vendor Concentration

NovaTech currently routes 100% of Project Atlas LLM inference through the Anthropic API (Claude claude-sonnet-4-6). An Anthropic outage would cause complete Project Atlas service unavailability. Mitigation: integrate a secondary LLM provider (OpenAI or AWS Bedrock) with automatic failover via

LiteLLM router. Target: Q3 2026.